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# Approaches to Synthetic Data Generation: Insights from Luxembourg's Census Data[1]

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## Abstract

We implement a synthetic data generation framework on a pseudonymized subset of the 2021 Census data for Luxembourg. Focusing on seven categorical variables—including ordered age and education—we drop unique records upfront to mitigate the risk of singling out. Synthetic data are produced via the CART method in the synthpop package. Utility is measured using the propensity score mean-squared error metrics and regression analyses, confirming that key statistical relationships are retained. In addition, privacy is further reinforced by aggregating numerical values (e.g., grouping ages) in combination with the inherent randomness of the data generation process, thereby mitigating disclosure risks related to linkability and inference. By addressing the three core EU guidance concerns—singling out, linkability, and inference—our methodology meets standards for anonymization and statistical disclosure control, keeping disclosure risk minimal while maintaining data utility. In this paper, we present our approach and highlight lessons learned.

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## Abstract

We implement a synthetic data generation framework on a pseudonymized subset of the 2021 Census data for Luxembourg. Focusing on seven categorical variables—including ordered age and education—we drop unique records upfront to mitigate the risk of singling out. Synthetic data are produced via the CART method in the synthpop package. Utility is measured using the propensity score mean-squared error metrics and regression analyses, confirming that key statistical relationships are retained. In addition, privacy is further reinforced by aggregating numerical values (e.g., grouping ages) in combination with the inherent randomness of the data generation process, thereby mitigating disclosure risks related to linkability and inference. By addressing the three core EU guidance concerns—singling out, linkability, and inference—our methodology meets standards for anonymization and statistical disclosure control, keeping disclosure risk minimal while maintaining data utility. In this paper, we present our approach and highlight lessons learned.

## I. Introduction

The creation of synthetic census datasets can help balance data privacy and accessibility. It allows researchers and analysts to explore meaningful patterns while keeping individual

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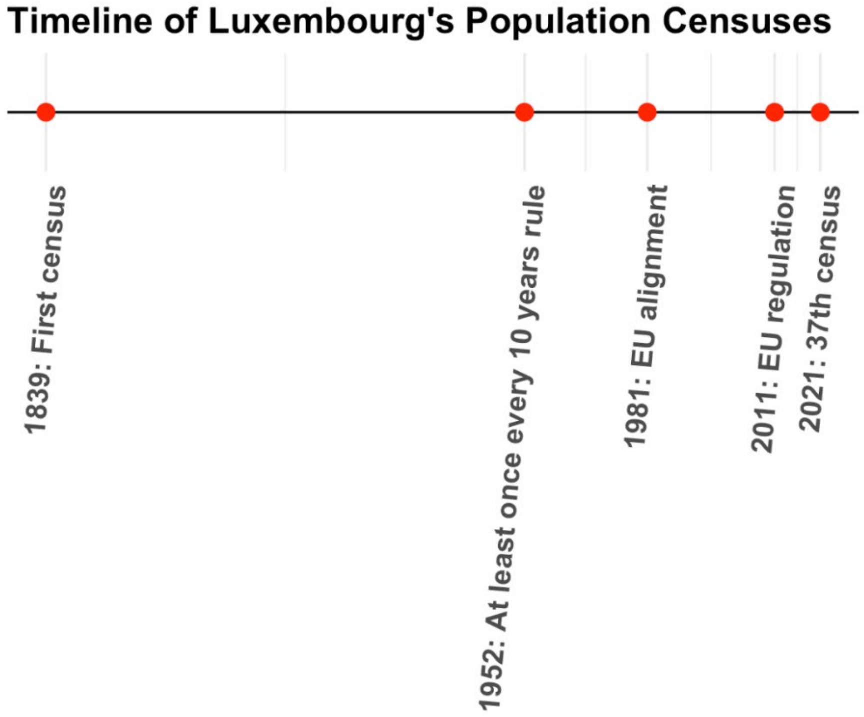
confidentiality risks minimal. Building on the United Nations Economic Commission for Europe (UECE) 2019 guide, this report examines various approaches for generating synthetic data, particularly in census applications. Additionally, it reviews a commonly used tool, *Synthpop*, in the field and how it relates to the core methods. The report wraps up with recommendations for generating synthetic census data for our use case of STATEC's 2021 census.

The 2021 Population Census of Luxembourg, conducted between November 8 and December 5, 2021, marked a significant advancement in data collection methodologies. For the first time, Luxembourg adopted a "combined" census approach that integrated data from both questionnaires and administrative registers. This new approach was designed to reduce the burden on citizens while enhancing the accuracy of the collected data. Notably, 48% of respondents completed the census online, demonstrating a significant shift towards digital participation.

Historically, population censuses are among the oldest and most widespread statistical operations, playing a crucial role in documenting a country's socio-economic structure. In Luxembourg, the first census after gaining independence was conducted in 1839. Subsequent censuses took place at irregular intervals until 1952, when legislation established a maximum interval of ten years between censuses. The 1981 census was the first to be organized under a European Directive, and by 2011, the process was governed by European Regulation, with that year's census taking place on February 1. The 2021 census represents the 37th since Luxembourg's independence, continuing this important tradition of gathering essential demographic data. Figure 1 illustrates the historical evolution of censuses in Luxembourg.

Figure 1: Historical evolution of Luxembourg’s censuses




The 2021 census recorded a total resident population of 643,941 as of November 8, 2021. This figure indicates a significant demographic growth of 25.7% (an increase of 131,588 people) compared to the 2011 census, which recorded 512,353 inhabitants. This substantial increase underscores Luxembourg's dynamic population changes over the past decade.[4]

In terms of age demographics, the population's average age in 2021 was 39.7 years, reflecting an aging trend compared to previous years, more precisely, 1 year older than in 2011. The young-age dependency ratio, representing the number of individuals aged 0-14 per 100 people aged 15-64, decreased from 25.2% in 2011 to 23.0% in 2021. Conversely, the old-age dependency ratio, indicating the number of individuals aged 65 and over per 100 people aged 15-64, increased from 20.4% in 2011 to 21.2% in 2021. These shifts highlight the evolving age structure of Luxembourg's population.

[4] https://statistiques.public.lu/en/publications/series/rp-2021/2023/rp21-02-23.html?utm_source=chatgpt.com

Based on these insights, we place more emphasis in our current project on key demographic variables: individual age (capped at 100 years), sex, economic activity status, education, occupation (CITP1), industry sector (NACE), and main language. These variables are helpful for understanding the population's composition and informing policy decisions. For instance, data on age and sex distributions help in planning for healthcare and educational services, while information on economic activity and education levels assists in labor market analyses. The census includes many other variables that could be leveraged. By demonstrating the feasibility for generating synthetic data from census data, this project paves the way for applying the method to a wider set of variables.

# II. Methods for Synthetic Data Generation

## A. Sequential Modeling Method (Fully Conditional Specification / SRMI)

Fully Conditional Specification (FCS) was originally developed within a data imputation context (Van Buuren et al. 2006). It is also known as Sequential Regression Multivariate Imputation (SRMI), a term introduced by Raghunathan et al. (2001), in the data imputation context.[5] In the synthetic data literature, Drechsler (2011) uses these two terms and continue with the FCS as the main term.

FCS has become a popular method for generating synthetic data. Specifically, it models each variable conditionally on previously synthesized variables in a sequential manner. This approach can be represented by factorizing the joint probability distribution into a series of conditional and univariate distributions:

$$f(X_1, X_2, \dots, X_p) = f(X_1) \times f(X_2 \mid X_1) \times \dots \times f(X_p \mid X_1, X_2, \dots, X_{p-1})$$

This method is flexible and well-suited for datasets with mixed variable types, such as continuous, binary, and categorical variables. It allows the estimation of each variable sequentially, ensuring marginal and joint distributions are preserved. Owing to these

[5] https://stefvanbuuren.name/fimd/sec-FCS.html

advantages, several National Statistical Offices (NSOs), such as Statistics Canada and Statistics New Zealand, use Synthpop (Nowok et al., 2016) to implement the FCS approach when generating synthetic data.[6]

Like any approach, this method offers both advantages and challenges. On the one hand, it preserves marginal and joint distributions effectively, allowing synthetic data to closely resemble real data and maintain simple variable relationships. On the other hand, because it proceeds sequentially, the method may struggle with capturing complex, non-linear interactions or hierarchical structures. It tends to be most effective with datasets that have relatively straightforward interdependencies. Nonetheless, the use of Classification and Regression Trees (CART), a non-parametric method, within Synthpop can help mitigate these limitations, as discussed further in Section V.

## B. Simulated Data Method

The Simulated Data Method generates synthetic data using assumed statistical distributions rather than relying on observed microdata. In this approach, variables are simulated from known or estimated distributions (e.g., normal or Poisson distributions) with fixed parameters such as the mean and variance that approximate the characteristics of real data (Drechsler, 2011).

Although this method enables the rapid creation of large-scale synthetic datasets and proves particularly useful when real data are unavailable or incomplete, it does not capture the complex multivariate relationships among variables. Consequently, while effective for preliminary model testing or situations where detailed data interdependencies are not critical, the method is less suitable when preserving complex associations is essential.

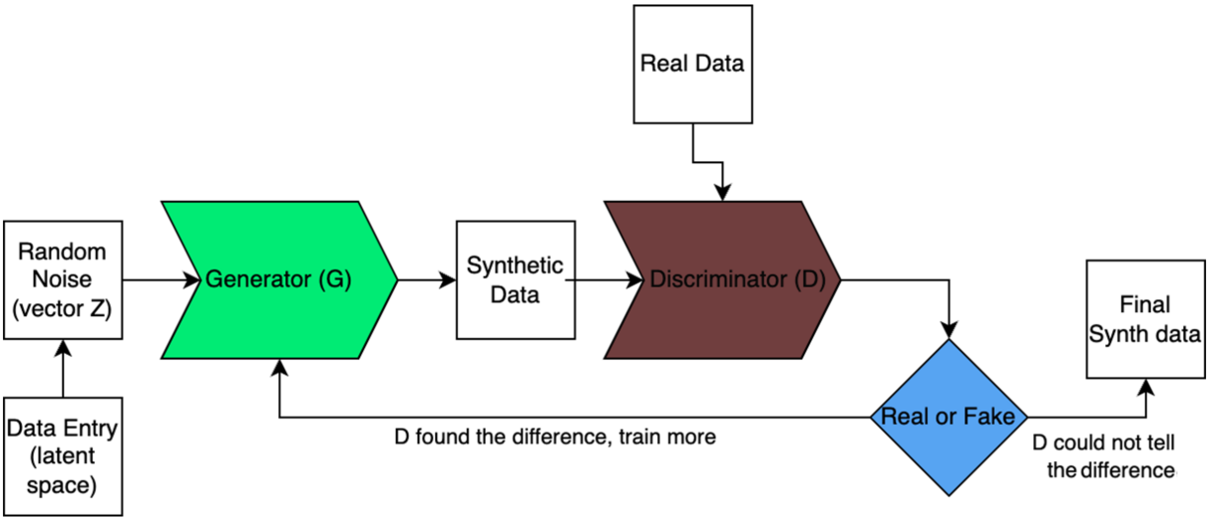
## C. Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) are deep learning architectures employed for generating synthetic data. Each GAN model consists of two competing neural networks—a generator, which creates synthetic data, and a discriminator, which attempts to distinguish the

[6] https://pages.nist.gov/HLG-MOS_Synthetic_Data_Test_Drive/guide.html#sat_gpa_synth_cart; https://www150.statcan.gc.ca/n1/en/pub/11-522-x/2021001/article/00026-eng.pdf?st=ZeNNhRtY;

synthetic data from real data. By iterating through this adversarial process, GANs can capture complex, non-linear, and high-dimensional relationships within the dataset. In practice, the Office for National Statistics (ONS) has utilized this approach to generate synthetic datasets for testing the Census 2021 system (UNECE, 2019). Figure 2 illustrates the fundamental GAN workflow, highlighting how the two networks interact during training.

Figure 2: Workflow of GANS




While GANs excel at producing large and high-dimensional synthetic datasets and can capture relationships that are challenging for traditional methods, they are also computationally intensive and often difficult to implement. Training GANs requires specialized expertise and can become unstable, particularly when working with smaller datasets. Additionally, we believe that a reverse engineering might be easier here given that a pure machine learning technique, unless some privacy preserving techniques are added on top of the synthetically generated data. However, this might affect the utility gained so it has to be done cautiously.

### D. Hierarchical Bayesian Models (HB)

Hierarchical Bayesian Models (HB) use probabilistic modeling with Bayesian inference to generate synthetic data. HB models are particularly useful when dealing with hierarchical or nested data, such as individuals within households and households within regions, which is typical of census data.

By explicitly modeling variability both within and between groups, HB approaches tend to preserve the complex dependencies inherent in such data (Drechsler & Reiter, 2010, Young, Graham, & Penny, 2009). However, the computational demands of HB models can be substantial, making them challenging to implement on large census datasets.

# III. Recommended Method and Tool for Synthetic Data Generation

## A. Recommended method (Fully Conditional Specification)

Among the methods discussed, Fully Conditional Specification (FCS) emerges as the most practical and flexible approach for generating synthetic census data. It is a robust method, particularly effective for creating public-use microdata files while balancing data utility and confidentiality. By synthesizing each variable conditionally based on previously synthesized variables, FCS preserves both the marginal and most related joint distributions of the original data. This sequential modeling process is particularly effective for datasets containing a mix of continuous and categorical variables, ensuring that the synthetic data closely resembles the original dataset in structure and relationships.

FCS is adopted by National Statistical Offices (NSOs) such as Statistics Canada and Statistics New Zealand. Its adaptability and reliability have made it the preferred method for generating high-quality synthetic datasets for census applications in some NSOs. The method is especially suited for datasets where the preservation of simple and complex variable relationships is critical to ensuring analytical utility and maintaining confidentiality.

While alternative approaches, such as Hierarchical Bayesian Models (HB) or Generative Adversarial Networks (GANs), offer advanced capabilities for specific cases like hierarchical data or non-linear modeling, FCS remains the most practical and easy to implement for census data. Techniques such as Classification and Regression Trees (CART) within the FCS framework, as we will see in the subsection III.B.

In conclusion, FCS is the most appropriate method for generating synthetic census data when considering the practicality as well as no deep domain knowledge as a pre-requisite. Its ability to preserve critical data relationships, coupled with its flexibility and widespread adoption by leading NSOs, makes it the ideal choice for projects requiring both high utility and confidentiality.

## B. Recommended tool (Synthpop)

Following the recommendation of Fully Conditional Specification (FCS) as the most suitable method for generating synthetic census data, Synthpop emerges as an excellent tool to operationalize this approach. Synthpop, widely adopted by National Statistical Offices (NSOs), offers flexibility and robust functionality for creating synthetic datasets that closely resemble original data while preserving both marginal and joint relationships.

### i) Key Features of Synthpop

In Synthpop, the synthesis method leverages techniques such as the Classification and Regression Trees (CART) for variables with predictors (Nowok et al., 2016). CART is particularly effective at handling both continuous and categorical variables by capturing complex, non-linear relationships. It helps generate realistic synthetic data that maintains the original data's essential patterns and relationships.[7]

Synthpop generates synthetic data in a carefully ordered, one variable at a time fashion. The first variable, chosen for its simplicity or independence, is drawn unconditionally by sampling directly from its observed marginal distribution. Each subsequent variable is then synthesized conditionally, using the values of all previously generated variables as predictors. This approach ensures that both the marginal distribution of each variable and the full pattern of joint relationships across variables are faithfully reproduced in the synthetic dataset.

Synthpop offers extensive customization of this sequential process. Through the visit.sequence argument, users can tailor the exact order in which variables are synthesized, rather than relying on the dataset's column order. This allows particularly important predictors to be generated early or late, as needed. By specifying the number of imputations (m), analysts can create multiple independent synthetic replicates, refining estimates of utility and providing a basis for quantifying uncertainty. The predictor.matrix feature gives granular control over which variables are used as inputs for each synthesis model, allowing the omission of weak or extraneous predictors and the inclusion of only the most relevant covariates. This combination of sequential modeling and user-driven options makes Synthpop a powerful, flexible tool for producing high-utility synthetic data.

[7] Of note, other alternative synthesis methods that Synthpop allows users to choose from can be parametric approaches or advanced techniques like Random Forests, depending on the dataset's complexity and the relationships being modeled.

### ii) Advantages of Synthpop

By using the CART-based FCS method, Synthpop generates synthetic datasets that retain the overall patterns (i.e., marginal distributions of variables) and the relationships between variables (i.e., conditional distributions). This ensures that the synthetic data closely reflects the important structures and interactions within the original dataset.

Moreover, Synthpop accommodates a wide range of synthesis needs by enabling users to switch between simpler parametric methods and advanced machine learning techniques like Random Forests. This flexibility makes it suitable for datasets of varying complexity and relationships.

### iii) CART in Action: Synthesizing Categorical and Continuous Variables

#### (a) Categorical Variables

CART models a categorical target by recursively partitioning the data into increasingly homogeneous groups. At each split, it evaluates all candidate predictors (e.g., age, income, education) and chooses the split, whether a numerical threshold or a subset of factor levels, that minimizes impurity. This data-driven process can, for instance, reveal that individuals under 25 are predominantly single, while those aged 25–40 are mostly married, without any predefined thresholds. After the initial split, CART examines each resulting subgroup in turn, determining whether further splits (perhaps based on income or education within the 25–40 age group) can improve purity. Splitting continues until a stopping rule is met, typically a minimum node size or a minimum improvement in impurity, ensuring that each terminal node contains observations that are as similar as possible in their outcome category. This approach ensures that the final tree captures the key patterns and interactions in the original data, which synthpop then uses to generate synthetic outcomes that faithfully reproduce those relationships.

#### (b) Continuous Variables:

When applied to continuous variables, CART acts as a regression tool that divides the dataset into segments with minimal variability in the target variable, such as income. The goal is to create subgroups where the value of the target variable is as consistent as possible, ensuring accurate predictions within each segment. CART achieves this by evaluating all possible predictors and finding splits that minimize the sum of squared errors (SSE), a measure used to assess the consistency of values within each group.

For example, if the target variable is income, and the predictors include age and education level, CART might find that individuals under 30 years old have relatively low and consistent income, creating one segment for this group. Those between 30 and 50 with a college degree might see a significant increase in income, leading CART to make a separate split. For individuals over 50, income might stabilize or vary less significantly, resulting in another distinct group. These splits are based on actual data patterns rather than arbitrary thresholds, allowing CART to adapt to the unique characteristics of the dataset.

CART continues this process, refining the splits iteratively until it reaches a point where further splits do not significantly reduce variance or meet other stopping conditions, like a minimum node size or a minimum improvement in impurity. The final output is a decision tree where each "leaf" represents a group of observations with relatively uniform values, allowing the predicted outcomes to be accurate. The predicted income for each group is often calculated as the mean or median of that group, ensuring that the model effectively captures the data's inherent relationships.

### iv) Technical Aspects of CART Splitting Criteria

In Synthpop, the cart method delegates to the rpart package to build classification and regression trees.[8] For classification, rpart by default uses the Gini impurity criterion at each node to choose splits that most reduce class mixing. For regression, it minimizes the sum of squared errors (SSE) within each child node. At every step, CART tests all possible thresholds (for numeric predictors) or level-partitions (for factors) and selects the split with the greatest purity gain (classification) or SSE reduction (regression). This data-driven process automatically uncovers the most important predictors and their cut-points (e.g. “age < 25” vs. “age ≥ 25” when predicting marital status) without any manual thresholds. Splitting continues until the complexity parameter (cp) and minimum bucket size (minbucket) stopping rules are met, ensuring each terminal node is as homogeneous as the data support. Synthpop then samples synthetic values by dropping each record down these trees and drawing from the observed outcomes in its terminal leaf—thereby preserving both marginal and joint patterns.

[8] https://cran.r-project.org/web/packages/rpart/index.html

### v) Recommended number of variables to generate

For the purpose of our exercise, we focus on a subset of seven categorical variables, two ordered and five nominal, to maintain tractability in our modeling efforts and to gain insights into how CART handles mixed variable types. This targeted selection allows us to isolate and diagnose utility issues, such as unusually high Propensity Mean Square Error (S_pMSE) scores, without the added complexity and computational burden of a larger set of variables. Once the subset of variables workflow is optimized and reliable, we can confidently expand the model. This approach mirrors the successful scaling of Synthpop to 35 variables by Nowok et al. (2016), ensuring that our tuning strategies, such as the order of synthesis and tree parameters are robust and generalizable. By adopting this stepwise methodology, we balance computational efficiency, interpretability, and utility, ensuring that each additional variable incrementally enhances an already validated pipeline.

# IV. Disclosure risk and privacy preserving techniques

## A. Legal background

The General Data Protection Regulation (GDPR) regulates the processing of personal data in the European Economic Area. Since techniques of synthetic data creation have been recognized as a way to anonymize personal information (AEDP, ICO, Datatilsynet) and as the project involves the creation of synthetic data from real word data, it is necessary to pay attention to some data protection law requirements and concepts. In particular, it is necessary to consider the concepts of personal data, of anonymous data, and identifiability of data subjects.

The GDPR defines personal data as any information relating to an identified or identifiable natural person. A person is considered identifiable when it can be identified directly or indirectly via an identifier or any other aspect related to the identity of the person, such as an ID number, an online identifier or factors related to the physical, economic or social life of an individual.

Although the GDPR sets down a broad definition of personal data, recital (26) identifies certain limits for determining in practice when a person is identifiable. Namely, all the means that are reasonably likely to be used, by the controller or by any other person, to make an individual identifiable should be taken into account when deciding whether a piece of information is personal data. Moreover, recital (26) advises to consider objective factors, such as the time, cost and technology available for determining if these reasonable means exist or not.

Therefore, on the other hand, the information that is rendered anonymous (anonymized) in such a way that a data subject is not, or no longer, identifiable shall not fall under the scope of the GDPR. Nonetheless, the GDPR does not provide any further clarification about how to perform the anonymization process. The GDPR focuses on the outcome of the anonymization process, rather than on the process itself. Hence, in case a data controller engages in an anonymization activity, he should be able to demonstrate that individuals are no longer identifiable from anonymized data. In other words, even after applying anonymization techniques there is a presumption that the data are still personal data unless the data controller can demonstrate that individuals are no longer identifiable.

The Article 29 Data Protection Working Party enacted an opinion in 2014 on anonymization techniques. In the document, the WP29 carries out an analysis of anonymization from two different perspectives. First, from the legal point of view, the WP29 tries to clarify when the outcome of the anonymization process can be considered acceptable. Second, from the technical point of view, the WP29 lists some technical measures that can be applied for performing the anonymization process. Focusing on the legal aspects, essentially, the WP29 recognizes that anonymization carries a risk factor, which is the risk of re-identification of the individual. When assessing this risk, according to the WP29, all the means likely usable to re-identify such an individual should be taken into account to understand whether the risk exists or not. Moreover, the WP29 identifies three potential re-identification scenarios in the anonymized dataset: 1) risk that the individual's records can still be isolated within the anonymized dataset (singling out); 2) risk that at least two records of the same individual are linked together within the same dataset or two different datasets (linkability); 3) risk that information about an individual are inferred from a set of other attributes (inference).

To sum up, when anonymization techniques are used, a risk assessment concerning the risk for an individual of being re-identified should be carried out. The risk assessment should take into account the means reasonably likely to be used by any person for either singling out an individual, link information about such an individual, or infer information about him. It has been argued, quite unanimously, in literature (M. Finck, F. Pallas; S. Stalla-Bourdillon, A. Knight; K. El Emam, C. Alvarez) that the re-identification risk can never be zero. First of all, because a zero-risk approach clashes with the concept of risk itself and consequently with the general GDPR approach to legal compliance. Moreover, the GDPR itself seems to introduce a "reasonable" approach when mentioning, in recital (26), an assessment of the means likely to be used to identify data subjects. Finally, the GDPR requires that the risk assessment should be carried out from the point of view of any person potentially able to re-identify the data subject, not just from the data controller perspective. Enlarging the perspective of any person who can re-identify the individuals inevitably entails the admission that the risk of re-identification should be context-oriented and prone to changes over time. Indeed, the data controller cannot be sure of the means, at the disposal of potential third parties, that can be used to re-identify individuals, including technological advancements. All the above considerations inevitably lead to the conclusion that the re-identification risk assessment cannot follow a zero-risk approach, but it should rather aim at an acceptable risk of re-identification, instead, in consideration also of the context within which the data are disclosed. On the other hand, this is not to say that the risk of re-identification should be disregarded, as it has been demonstrated that poorly anonymized datasets after being publicly released have been subject to re-identification of the individuals (see for example, L. Sweeny; A. Narayanan,V. Shmatikov; Y. Montjoye et al; J. Su et al.; L. Rocher, J.M. Hendrickx, Y. de Montjoye).

## B. Disclosure risk

In this session we explain how the three risks of re-identification mentioned above, singling out, linkability, and inference, have been addressed while creating the synthetic data in order to achieve a risk of re-identification that is considered acceptable.

### 1. Singling Out

As a general rule, the risk of re-identifiability based on singling out arises whenever a dataset contains information relating to a natural person with a unique combination of attributes. This is also sometimes referred to as a "small cell" problem. Given that the initial dataset in question describes (almost) the entire Luxembourgish population, at a minimum, the data subjects with the unique characteristics might be able to readily identify themselves. Likewise, any viewer of the dataset would have little trouble identifying their acquaintances who happen to have a unique combination of characteristics within the dataset. In many cases, it would also be possible for the viewer to identify individuals other than one's acquaintances, using little-to-moderate amount of effort (e.g., performing an online search to find social media profiles of people with matching characteristics). In the context of synthetic data, this risk may however be lower compared to other forms of (allegedly) anonymized data, because the generation of synthetic data results in alterations or differences in the data points in a record in comparison with the source dataset in addition to the removal of identifiers. In addition, we aggregate by age groups besides the inherited reshuffling and randomness added by the synthetic data generation that are considered privacy preserving techniques to anonymize data (Fung, Wang, Fu, & Yu, 2010).

We address singling-out risk assessment by measuring the identity disclosure risk which refers to the ability to identify individuals in the data from a set of known characteristics that are called keys. The evaluation of the identity disclosure measure on the original data gives the information about the unique combinations of the keys' values before ingesting into the synthetization process. Therefore, we evaluate the percentage of records that are distinguishable by looking at the uniqueness of the key's variables in the original and in the synthetic dataset. Upon synthetization, we expect that this percentage is smaller since the synthetic data should preserve privacy of individual sensitive information more than the original dataset and therefore reduce the risk of singling-out.

### 2. Linkability

Linkability between datasets or other sources of information may give rise to re-identification risks in certain cases. It's important to highlight that linkability is not limited to establishing cross-references between two or more structured datasets. For instance, in the case of individuals within a dataset who have a unique combination of characteristics, re-identification attempts can be made by searching additional information online and linking the findings with the data subject's characteristics known based on the dataset (see section above "Singling Out").

Linkability is harder to achieve with synthetic data compared to data anonymised using other techniques, insofar as synthetic records do not correspond to real individuals. Therefore, there is no one-to-one mapping between synthetic records and real data subjects, making it fundamentally more difficult to reidentify anyone. Moreover, introducing randomness and reshuffling helps to even out rare or unique attribute combinations, which are often targeted in reidentification attacks.

Indeed, a simple one to one record merging failed to connect synthetic records to their original counter parts by using a set of several key variables highliting the difficulty of linking the synthetic data to original records from other data sources.

### 3. Inference

Inference has to do with deducing a new piece of information/attribute about the data subject based on the available information. Importantly, the ability to infer a new attribute, in and of itself, will typically not be sufficient to re-identify data subjects. To establish the data subject's identity, the newly derived attribute must be such that it would allow singling out an individual within the dataset, or the person attempting to re-identify the data subject would able to link the newly derived attribute with another piece of information, e.g., by cross-referencing it with another source of data. In other words, to determine whether the ability to infer a new piece of information about the data subject increases the risk of re-identification, inference should be considered in conjunction with "singling out" and "linkability".

The inference risk is relatively lower in synthetic data, because synthetic data records do not directly correspond to real individuals. Identity disclosure can even be considered less relevant for completely synthesized where when all values of all variables are replaced by synthetic values (Raab et all., 2025). Because synthetic records are not tied to real people, an attacker cannot confidently infer a real person's attributes based on what they see in the synthetic data alone. This inherent disconnect between synthetic records and real individuals significantly weakens the basis for valid inferences. However, inference risk is not entirely eliminated: attackers can still exploit the synthetic data to learn population-level or group-level insights, which may indirectly reveal sensitive information in certain cases. Moreover, if attackers possess strong auxiliary knowledge, they might use synthetic data to confirm or refine what they already suspect about someone, especially if the synthetic data closely mirrors the original data's distributions. Though starting from pseudonymized data besides using the anonymization techniques of micro-aggregation on age level and the randomness and reshufflings done by synthetic generation keep this inference at a minimum level, especially that all our variables are categorical and we end up with a lot of identical records in our data.

## C. Conclusion of the risk assessment

We employ techniques to keep the risk of reidentification as minimal as possible, given that achieving zero risk while preserving acceptable data utility is considered a not realistic exercise. Our strategy consists of several privacy layers. First, we start with a pseudonymized set of variables, meaning that there is no direct identifiable variable. Second, we eliminate replicated unique records. These are records that are (i) unique in the original data and (ii) that have been generated in the synthetic data in a unique way. After the synthetic data generation process, we identify such cases and eliminate such records. Third, we use age categories instead of a continuous variable, which serves as an aggregation privacy-preserving technique, and this even amplified in our case of all variables being categorical. Fourth, the CART method applies additional privacy techniques, namely randomness and reshuffling, to capture the whole structure of the data.

By addressing the three core EU guidance concerns, singling out, linkability, and inference, our methodology aims to meet standards for anonymization and statistical disclosure control, maintaining minimal disclosure risk while preserving data utility, as much as possible.

Furthermore, to quantify the disclosure risk, we exploit the basic function disclosure provided by Synthpop to compare the identity and attribute disclosure risk inside the original dataset with respect to its synthetic version. We expect that the synthetic dataset will be more protected from a privacy point of view, and we demonstrate this intuitive assumption by calculating and relating the appropriate disclosure measures.

To balance utility and privacy, we first define the quasi-identifiers (called keys in the results) as the sensitive information that can be potentially available to an intruder to identify one individual profile. The rest of the variables are considered target variables for studying attribute disclosure in the context of inference. Therefore, these identified keys might be expected to be known to an intruder who believes he has access and can query real data instead of the actual synthetic data he possesses. We compare similar disclosure measures applied to original and synthetic data to quantify the level of additional protection reached.

Finally, we remind that this risk assessment should be reviewed periodically, especially in case of public disclosure of the synthetic data, or in case of other relevant changes in the synthetic data generation methods, as the risk might change over the time. To assess the achievement of legal compliance a case-by-case analysis is always necessary from the data controller. We further detail our approach and highlight lessons learned, demonstrating the effectiveness of our privacy-preserving techniques in section VI.

# V. Data Processing and Validation

First, we start by selecting seven categorical variables of interest, namely: age categories (ordered), sex, education (ordered), job status, occupation, industry, and master language spoken. We retrieved these variables from the original raw data following STATEC and EUROSTAT specific codes in form of JSON files. Second, we apply validation rules to make the dataset more coherent maintaining real feature relationships. For example, education and economic activity statuses are adjusted based on age groups. Third, the ordinal variables age and education, are ordered appropriately. Consequently, the education category “Not stated (of the persons aged 15 years or over)” is declared as missing value to not bias the ordered values.

# VI. Synthetic Data Generation Results and Necessary Tests:

## A. Utility measures

The main objective of this investigation is to find a reliable method and tool that allow us to generate synthetic data with an acceptable utility-privacy tradeoff. Since the original data are personal microdata we want to preserve privacy by keeping disclosure minimal. Therefore, we need objective utility and privacy metrics to guide us in finding desirable results.

One of the first ways to evaluate utility is by comparing univariate distributions between original and synthetic marginal distributions. This evaluation allows us to compare and visualize the goodness of the overlapping between the original univariate distributions and the corresponding artificial ones. Then we do the same for the joint distributions. The most common way to compare real and synthetic data in a uni and multivariate manner is to exploit a propensity score measure to test for distinguishability in utility between original and synethtic data. Propensity Mean Square Error (pMSE), is a broad utility measure that gauges overall distributional similarity between synthetic and original data. It is based on a propensity score model that attempts to distinguish original records from synthetic ones. In practice, one combines the original and synthetic data, labels them, and fits a classifier (often logistic regression or CART) to predict whether a record is “real” or “synthetic” using all available variables. pMSE is essentially the mean squared error of the predicted probabilities from this model.[9] If the synthetic data is indistinguishable from the real data (high utility), the classifier will perform no better than random guessing, yielding predicted probabilities around 0.5 for each record and a low pMSE value (close to the theoretical minimum). If the synthetic data differs significantly from the original (low utility), the classifier can separate them (predicted probabilities close to 0 or 1), resulting in a higher pMSE.

Although pMSE reflects overall fidelity, its raw scale is influenced by the complexity of the propensity score model (degree of freedom) and the sample size, making direct comparisons difficult. Standardized pMSE (S_pMSE) addresses this by dividing the observed pMSE by its expected value under perfect fidelity, essentially benchmarking the error against chance performance. An S_pMSE near one indicates that the synthetic data are about as similar to the original as a random sample, while larger values reveal systematic discrepancies. By adjusting for model flexibility and data volume, S_pMSE delivers a more consistent, interpretable utility score, enabling reliable comparisons across different syntheses and ensuring that improvements in synthetic-data generation are genuinely meaningful. Synthpop author recommend S_pMSE to be less than 10 to be acceptable and even better if the value is less than 3 (Raab et al., 2016),

### B. Privacy measures

To evaluate the risk associated with identity and attribute disclosure, we use the basic function *disclosure* that allows us to compute the identity disclosure for a set of quasi-identifiers (keys) and attribute disclosure for one variable defined as a target from the same set of keys. Specifically, the keys are chosen for being considered sensitive information in the possession of an intruder. Therefore, by setting up a worst-case disclosure scenario, we can control what available information could potentially open a way to learn more about the original data and expose real information to further disclosure. Obviously, we are conscious that the real disclosure risk from the use of synthetic data must be related to the context of its release (see Elliot, Mackey, and O'Hara (2020)). We assume that the intruder cannot easily access similar or complementary microdata and be able to perform a linkage attack on the synthetic data by using this additional information to enable matching the two datasets. That would require, behind the real feasibility of getting that supplementary information, applying advanced

[9] https://search.r-project.org/CRAN/refmans/synthpop/html/utility.gen.html

machine learning or deep learning techniques and spending a certain amount of time and computational resources to try retrieving original data without considering the required intellectual ability and acquirable knowledge needed to perform this introspection. Moreover, avoiding sharing the complete code, the exact implementation of the synthetization method, and hiding the explicit parameters setting can add another layer of protection against the malicious or prohibited use of synthetic data from being ingested as initial input into an inverse-engineering computational machine to guess the original real data. However, we assume that the code shared as it is cannot be effortlessly used to retrieve the real data because the initial microdata is in fact pseudonymized and aggregated upon being synthetized.

The provided measures are therefore intended to quantify the risk when only the synthetic data is under attack, and we want to evaluate its vulnerability based on some restricted knowledge on one or more individuals that an intruder have in the original dataset. The measures are computed for both original and synthetic datasets. Comparing them, we expect that the synthetic dataset should acquire lower disclosure risks even that a large dataset with few categorical features might have a disclosure risk already low.

To study identity disclosure risk which is related to a singling-out attack, we choose all the variables as quasi-identifiers (keys) and measure identity disclosure risk by computing the number of *replicated uniques* in the synthetic dataset in comparison with the number of records in the original data that have unique combinations of all key variables' values. This measure evaluates a record-level vulnerability for re-identification risk. We expect that this risk decreases with the size of the datasets since the chance that more people share the same sensitive combinations of values increases.

The attribute disclosure risk refers to the ability to discover some information on the target variables associated with the record in the original data from the keys. This measure is related to inference's weakness since partial real information on individuals is meant to evaluate the disclosure risk for the target variables that are not considered keys in possession by an intruder but attributes disclosable to him.

For testing linkability, final tests have been conducted between synthetic and original data using different sets of common identifiers. The failure to merge datasets confirms the effectiveness of privacy-preserving techniques in protecting against finding sensitive and disclosive similarity within the synthetic data and between both original and synthetic datasets.

## C. Results

Starting with already pseudonymized data and aggregated on age categories, we generate synthetic data using the CART method, with custom visit sequences to maintain correlations between variables. It is recommended that we start with the most independent variable, age, first (i.e., at left of the sequence), then most correlated are recommended to come after each other. Given that we are dealing with a mix of nominal and ordered variables, for simplicity, we transformed the categories into numeric values to see most associated variables. We find that job status, occupation, industry are associated. In addition to the visit sequence, we adjusted CART tree's complexity parameters to balance utility and privacy by the standardized propensity score mean-squared error. This metric compares between synthetic and original data to confirm that key statistical relationships are retained.

Following the testing of different scenarios, for our data we chose a visit sequence, tree parameters, and a set of validation rules. These parameters were identified after conducting a limited grid search strategy.

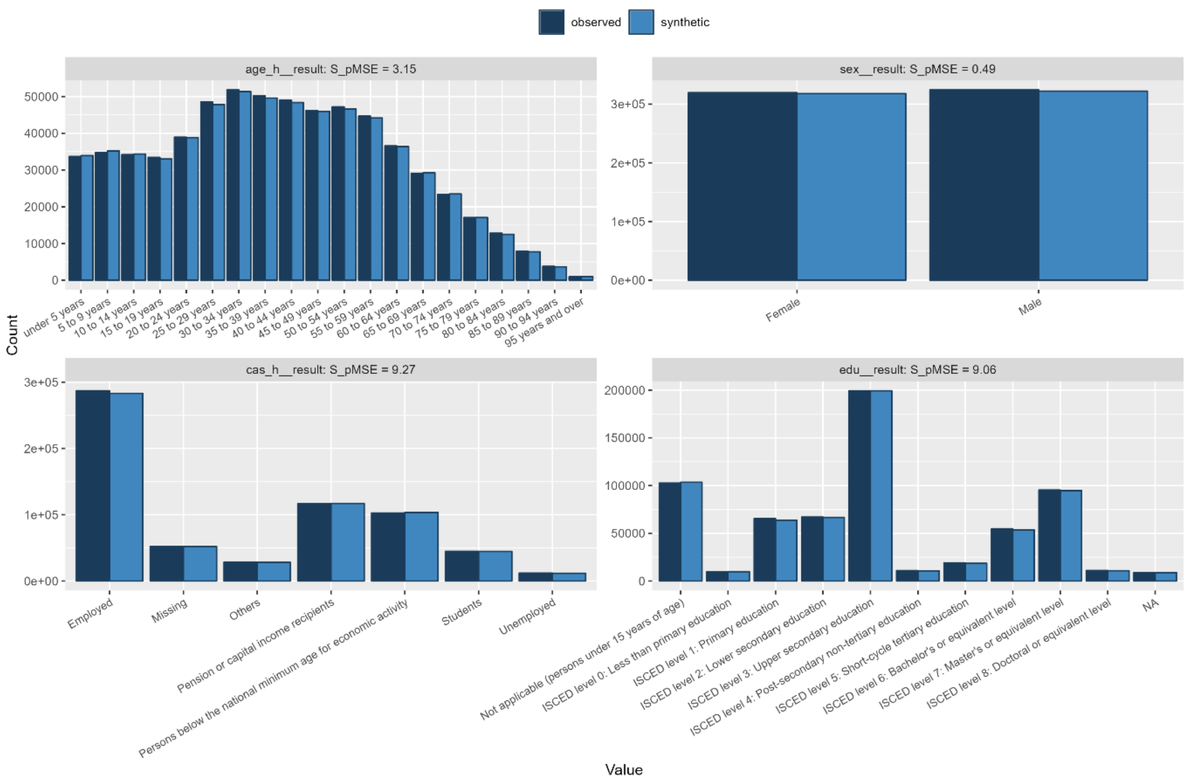
### 1. Utility results

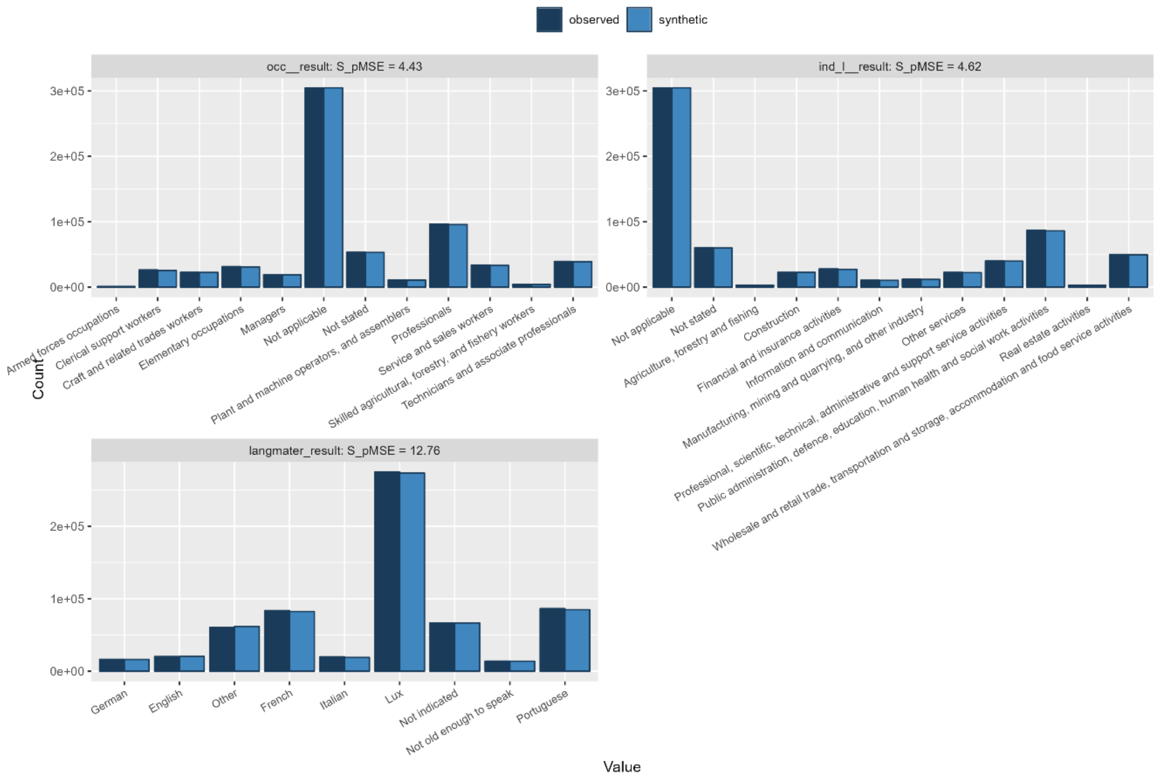
The univariate utility results reveal that synthpop's CART synthesis reproduces some marginals extremely well, sex has an S_pMSE of just 0.49, indicating nearly perfect agreement, while others show moderate to substantial discrepancies. Age (S_pMSE = 3.15) and the occupational sector (4.43) lie in the "acceptable but tunable" range, whereas education (9.06), household caseload (9.27) and industry (4.62) to the recommended upper limit (S_pMSE < 10). Language proficiency stands out with an S_pMSE of 12.76, flagging significant under- or over-representation in certain categories (see Table 1). The accompanying univariate plots illustrate these mismatches, for example, synthetic counts of "Not applicable" education or "Others" language groups deviate visibly from the originals. Taken together, these diagnostics suggest that while the overall algorithm captures most marginal distributions, targeted adjustments, such as refining the CART control parameters or reordering key predictors, are needed to bring all S_pMSE values below the preferred threshold (Figure 3).

Table 1: Univariate utility results

| Variable | pMSE | S_pMSE | df |
|---|---|---|---|
| age_h__result | 5.84 e-06 | 3.15 | 19 |
| sex__result | 0.05 e-06 | 0.49 | 1 |
| cas_h__result | 5.43 e-06 | 9.27 | 6 |
| edu__result | 8.85 e-06 | 9.06 | 10 |
| occ__result | 4.75 e-06 | 4.43 | 11 |
| ind_1__result | 4.96 e-06 | 4.62 | 11 |
| langmater_result | 9.97 e-06 | 12.76 | 8 |

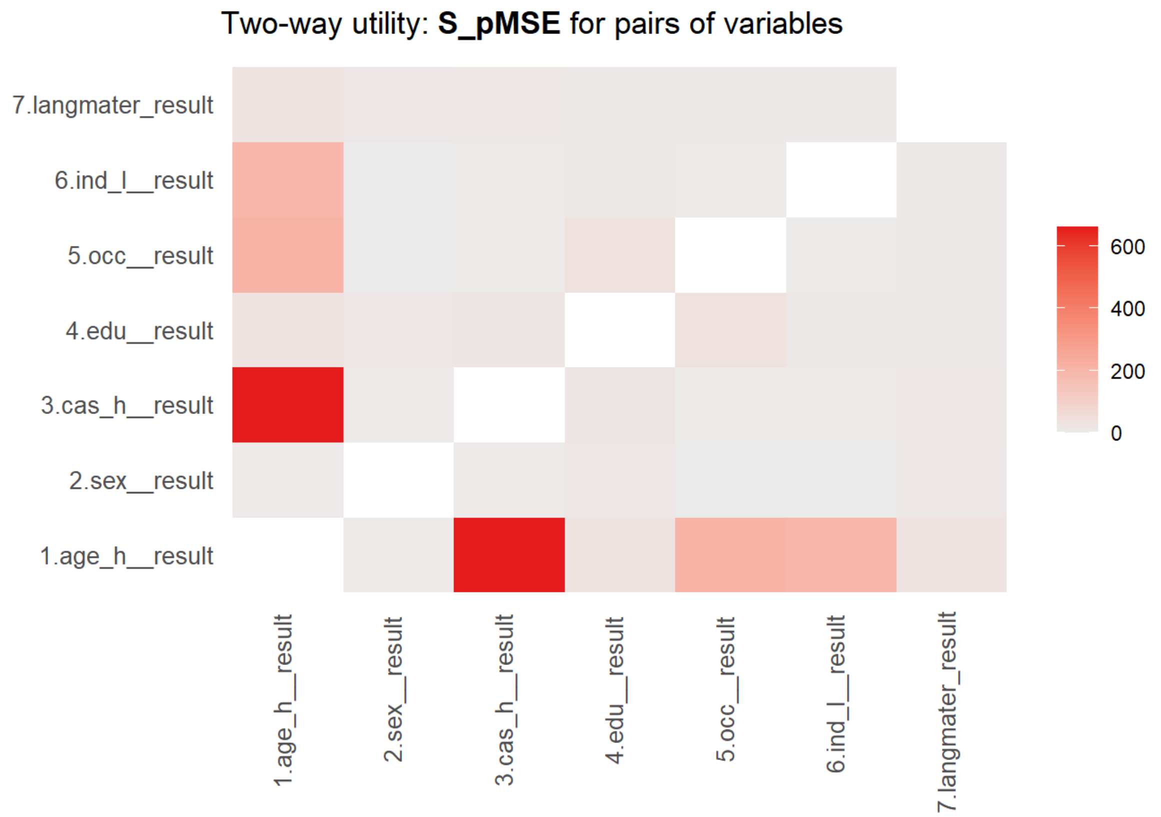
Figure 3: marginal (uni-variate) utilities






However, we also aim to preserve joint distributions, not just marginals. Our chosen visit sequence and CART tuning parameters were selected to balance marginal accuracy with multivariate fidelity, as evidenced by the two-way S_pMSE heatmaps (Figure 4).

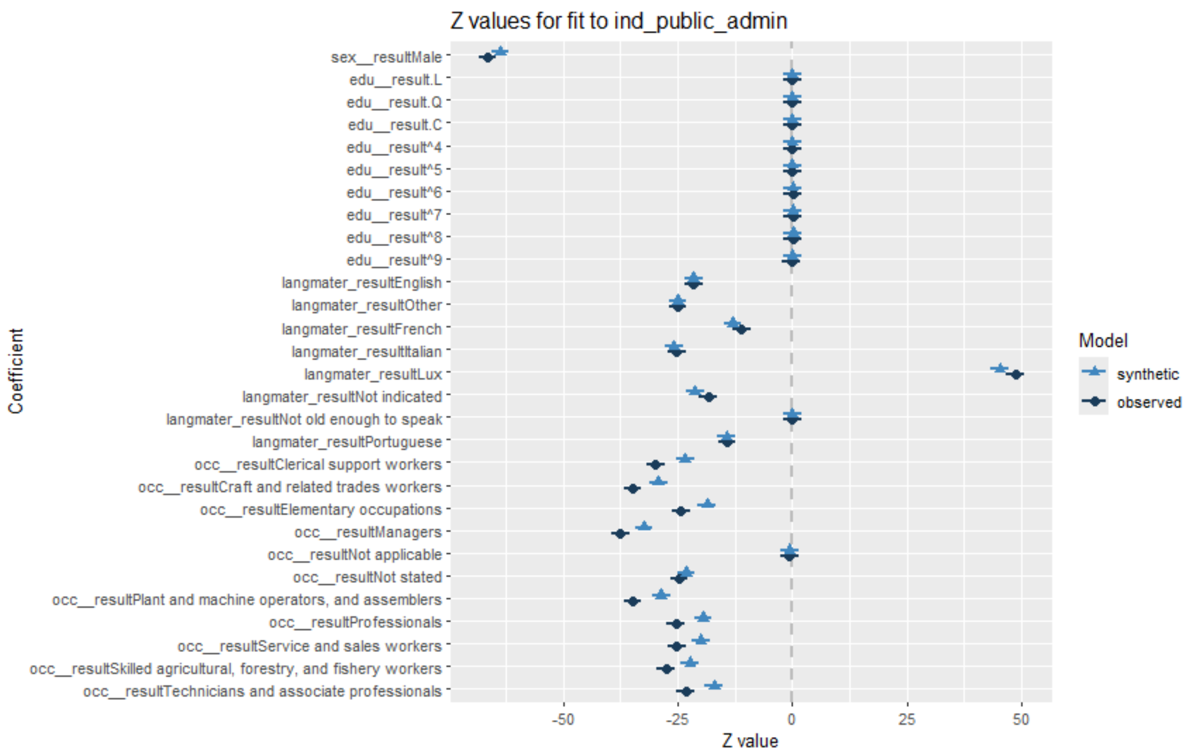
Figure 4: two-ways utility




To evaluate multivariate relationship preservation, we fit logistic regression models to both the original and synthetic datasets and compared three global indices. The model predicts whether an individual works in "Public administration, defence, education, human health and social work activities" (a binary outcome), using sex, education level, mother tongue (language), and occupation as predictors. The target variable takes the value 1 if an individual works in this NACE section, and 0 otherwise. Overall, the synthetic data coefficients largely retain the signs and ordering of the real data coefficients, indicating the main relationships are captured. (Please, see the full regression results in the Table I.1 in the APPENDIX I).

Figure 5 presents confidence intervals for the Z-statistic values of different model coefficients. In essence, Z-statistic value tells you how many standard errors the estimate is away from zero. The plot confirms that many relationships survive the synthesis process intact. Furthermore, for every predictor in the model, you can tell whether its statistical significance was preserved by looking at whether both the original data and synthetic data confidence intervals lie entirely on the same side of zero (significant in the same direction), or both straddle zero (non-significant in both). In the reported CI plot, every single synthetic data coefficient whisker sits on the same side of the zero line as the original data whisker. That means none of the predictors flip from significant to non-significant or vice versa (i.e., if the original model found a positive (or negative) effect that was statistically distinguishable from zero, the synthetic model does so too, and if the original found no clear effect, neither does the synthetic such as **education** predictors). In other words, the synthesis process has preserved the statistical significance (and direction) of every coefficient.

Figure 5: Confidence-interval plot of the coefficients' Z-statistic vales




In Table 2 we see that the average overlap of confidence intervals for each coefficient (0.44), indicating less than half the interval lengths overlap on average (detailed overlaps are shown in Table I.1 in APPENDIX I). Table 2 also confirms that the mean absolute standardized difference in coefficients (2.18), meaning that on average the synthetic estimate is about 2.18 standard errors away from the original estimate. The Mahalanobis-distance-based lack-of-fit ratio (2.84, $p < 0.001$) indicating that overall, the synthetic coefficients deviate more from the original than expected by chance.

In sum, the moderate CI overlap and modest coefficient differences indicate that many broad relationships are captured, but the significant Mahalanobis result shows that some joint correlations remain under-represented as we saw in Figure 4.

Table 2: Global indices from logistic regression

| **Metric** | **Value** |
|---|---|
| Mean CI overlap | 0.44 |
| Mean absolute std. coef. difference | 2.18 |
| Mahalanobis distance ratio (target = 1) | 2.84 |
| Lack-of-fit | 85.14 |
| p-value | <0.001 |

## 2. Privacy (risk assessment) tests

Privacy risk is assessed by comparing uniqueness rates and pattern overlaps between the original and synthetic datasets. Table 3 shows that in the real data, 18 161 records (2.82 %) have a unique combination of key variables, while in the synthetic data 13 406 records (2.08 %) are unique. Of those synthetic uniques, 3 559 records (0.55 % of all records) correspond to patterns that also occur in the original data, although none of those patterns were themselves unique there. Crucially, no record that was unique in the original data remains unique in the synthetic data (repU = 0.00 %). Although LNDS guidance recommends removing original unique records to eliminate any potential replication of unique patterns, our Replicated Uniques is already zero, and the overall uniqueness rate has dropped from 2.82 % to 2.08 %.
These results demonstrate that our synthetic dataset poses no greater, and in fact slightly lower, re-identification risk than the real data.

Table 3: Record Uniqueness

| **Category** | **Number** | **Total** | **Percent** |
|---|---|---|---|
| Original unique records | 18 161 | 643 941 | 2.82 % |
| Synthetic unique records | 13 406 | 643 941 | 2.08 % |
| Synthetic uniques in original | 3 559 | 643 941 | 0.55 % |
| Replicated uniques | 0 | 643 941 | 0.00 % |

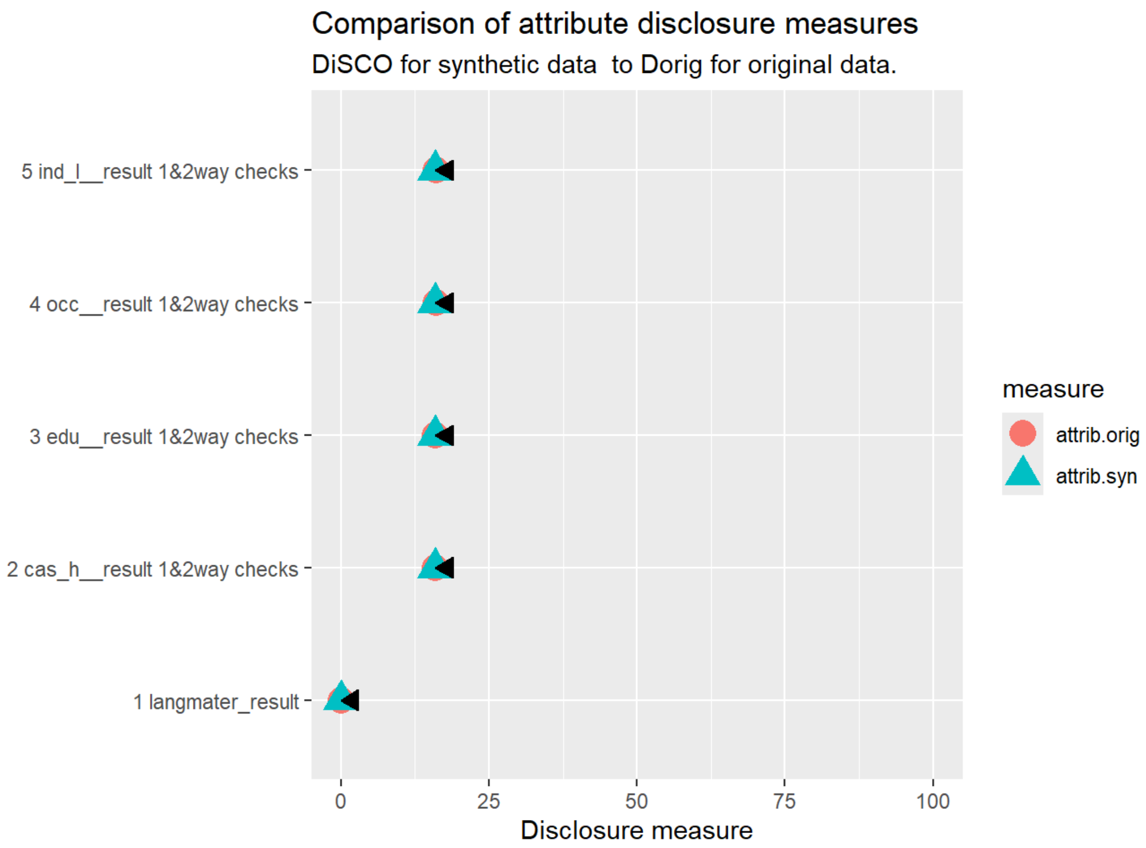
We applied attribute disclosure measures on target variables, which are attributes that an intruder tries to learn based on the keys' values he knows about an individual.
The measure *Dorig* (Orig. Attrib. in Table 4) counts the percentage of the records in the original dataset that identify a unique value of a target variable associated with a specific keys' values.
The measure *DiSCO* (Synth. Attrib. in Table 4) gives the proportion of the original records in which the target variable has the same level of the synthetic data. It quantifies the real disclosure on a value target variable when an intruder looks at the target variable on the synthetic profiles with the same combination of keys.
Across language, household caseload, education, occupation, and industry, the percentages are virtually identical (differences of only a few tenths of a percent), indicating that the synthetic dataset does not attribute target level differently from the original data for the chosen keys (age,sex). The attribute privacy disclosure risk remains the same as in original data (around 16% and 0 for langmater) since the synthetic data has a correct attribution to the target. We consider this as a result of the *rules* imposed for forcing the synthetization method to maintain realistic and meaningful relationships between the variables. These are specified in the *r_values* parameter. As long as we believe that the target variables do not expose sensitive information, we can accept the 16% attribute disclosure risk overall since trying to lower it further will worsen the dataset's utility and mindfulness.

Table 4 Attribute Disclosure (Multi-way Margins)

| **Variable** | **Orig. Attrib.** | **Synth. Attrib.** |
|---|---|---|
| langmater_result | 0.00 | 0.00 |
| cas_h__result (1&2-way) | 15.93 | 15.93 |
| edu__result (1&2-way) | 15.93 | 15.93 |
| occ__result (1&2-way) | 16.07 | 15.93 |
| ind_1__result (1&2-way) | 16.07 | 15.93 |

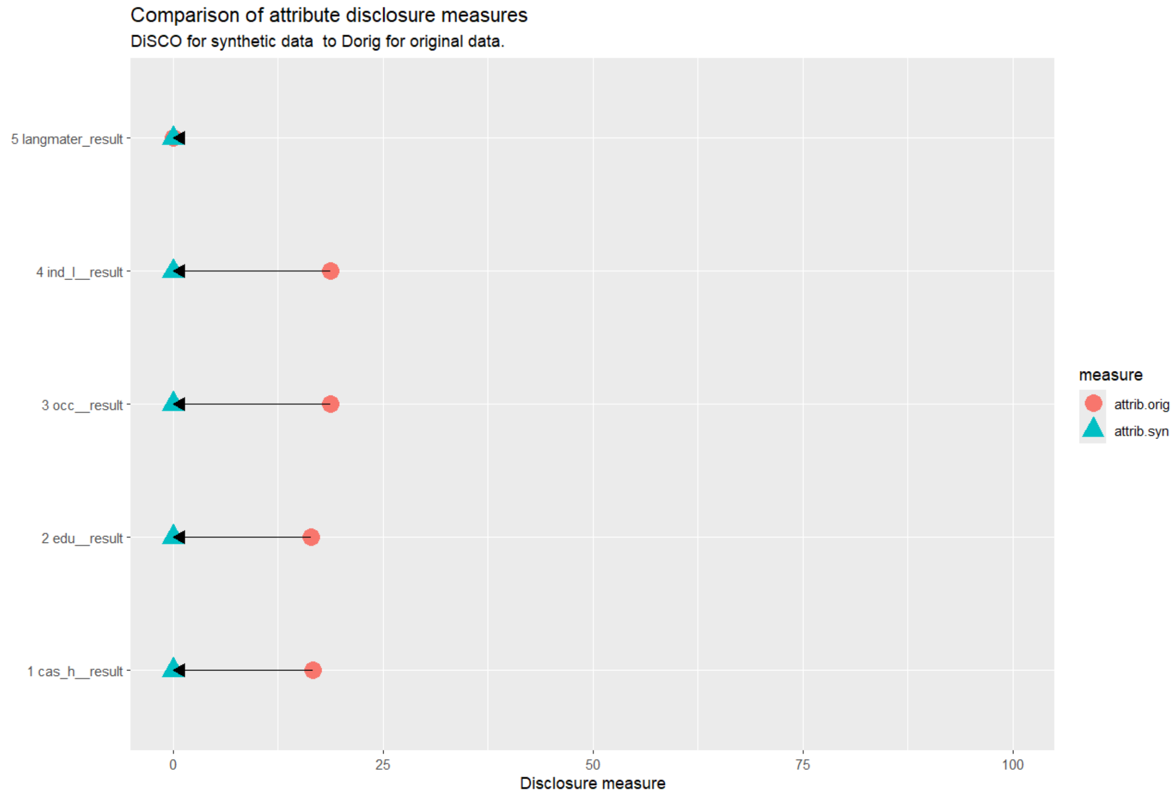
Further, Figure 6 shows each variable's most disclosive category (e.g. "Not applicable" for education) in each horizontal row. The red circle shows the original data percentage of unique value of each of these targets and the blue triangle shows the proportion of the original records that are disclosive in the synthetic dataset. The small black arrowheads mark the synthetic point over the red baseline to make any difference visually explicit. Across all five flagged variables language, household caseload, education, occupation, and industry the synthetic and original disclosure rates align almost perfectly, indicating that the synthesis process did not introduce any new attribute-level disclosure risk as well as any remarkable drop in the privacy protection for the target variables under investigations. Therefore, even the slight under-representation in the synthetic versions of some categories remains within a fraction of a percent, confirming that multi-way attribute disclosure is controlled to the same degree as in the original data

Figure 6: variables' most disclosive category




We also tested the scenario where "no rules" are imposed in the synthetic data generator, letting the method be capable of generating more variability. We find that privacy reaches values close to zeros for synthetic data, as you can see in Figure 7. This evidence demonstrates that the method itself can mask effectively attribute disclosure for the target variables when no restrictions are involved. On the other hand, we prefer to maintain these rules to preserve an acceptable level of utility, consistency and reliability.

Figure 7: Plot for attribute disclosure measures for target variables in the 'no rules' scenario.




Moreover, this free scenario does not flag variables. Therefore, flagged variables are a side-effect of the wanted rules. In a general context, those flagged variables would require deep inspections due to the possible disclosive risk of some levels of the target variables that are shared across many records (disclosure from 1-way). We claim that sensitivity on one specific target level is expected due to the rule specifications which relate keys and target specific values (e.g., education and economic activity statuses are adjusted based on age groups). We believe that we have reached an acceptable privacy-utility trade-off, more privacy protection would worsen utility, and privacy is preserved to the additional layer of protection we perform before synthesis.

Finally, for those variables flagged in the one-way margins, this shows the number of two-way variable pairs that still require inspection (i.e., where denominators are small or background knowledge suggests potential disclosure). Each affected variable here has exactly one such pair (See Table 5).

Table 5: Two-Way Check Counts

| Variable | Pairs Needing Check |
|---|---|
| cas_h__result | 1 |
| edu__result | 1 |
| occ__result | 1 |
| ind_1__result | 1 |

## VII. Discussion and conclusion

We start our synthetic generation from pseudonymized set of categorical variables and used individuals' age groups, instead of their actual age, as privacy preserving techniques by design. Then we employed a full-conditional (sequential) CART-based synthesis in R's synthpop package, fitting one tree per variable in a predetermined visit sequence. While this approach is highly effective at preserving marginal distributions, it may not reproduce the joint interactions fully at once. In our Luxembourg census example (two ordered, five nominal variables with 5–12 categories each), the combinatorial explosion of possible level combinations led to moderate one- and two-way utility results, albeit with identity risk reduced (unique key patterns fell from 2.8% to 2.1% for original and synthetic data, respectively, and 0% unique replicates from original to synthetic) and attribute-disclosure risk maintained at an acceptable ~16%.

Note that though we generate synthetic data, they may still be considered personal data under the GDPR if there's an unacceptable risk of re-identification for individuals included in the original census data (recital (26) GDPR and WP216 Article WP29). Recent research shows that even de-identified datasets can be highly vulnerable: with just 15 demographic attributes, 99.98% of individuals in U.S. data could be correctly re-identified (Rocher et al., 2019). While Rocher et al. (2019) did not focus on synthetic data specifically, the study highlights the risks of high-dimensional attribute combinations. To mitigate similar risks in synthetic data, we recommend limiting sequential CART synthesis to no more than 10 variables per batch. This helps balance data fidelity and privacy and avoids overfitting in univariate models that may not capture complex interactions.

By tuning tree complexity, enforcing validation rules, and carefully ordering variables, one can still reproduce the most critical one- and two-way relationships while keeping standardized pMSE values for marginals below 10. More precisely, each use case can prioritize few (3-5 out of 10) variables' marginal and joint distributions given that it is not possible to maintain a high utility for the whole set of variables.

Finally. This report shows that our approach's workflow yields a transparent, reproducible synthetic dataset that maintains robust privacy safeguards and preserves analytic utility for the most salient patterns in the original Luxembourg census data.

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## Annex I

Below is a side-by-side comparison of logistic regression coefficients (synthetic vs. observed) for the model predicting "Public administration … activities." Coefficient differences, standardized coefficient differences, and confidence-interval overlaps are all rounded to two decimal places

Table I.1: Regression analysis

| **Predictor** | **Synthetic** | **Observed** | **Diff** | **Std. Coef Diff** | **CI Overlap** |
|---|---|---|---|---|---|
| (Intercept) | 1.69 | 2.32 | −0.63 | −0.09 | 0.98 |
| sex__resultMale | −0.62 | −0.65 | 0.03 | 2.79 | 0.29 |
| edu__result.L | 0.47 | 0.44 | 0.03 | 0.00 | 1.00 |
| edu__result.Q | 0.55 | 0.63 | −0.08 | −0.00 | 1.00 |
| edu__result.C | 0.05 | 0.10 | −0.05 | −0.00 | 1.00 |
| edu__result^4 | 0.26 | 0.21 | 0.04 | 0.00 | 1.00 |
| edu__result^5 | 0.26 | 0.20 | 0.06 | 0.00 | 1.00 |
| edu__result^6 | 0.23 | 0.35 | −0.12 | −0.04 | 0.99 |
| edu__result^7 | 0.22 | 0.24 | −0.02 | −0.01 | 1.00 |
| edu__result^8 | 0.09 | 0.15 | −0.06 | −0.05 | 0.99 |
| edu__result^9 | −0.00 | −0.07 | 0.07 | 0.22 | 0.94 |
| language_english | −0.91 | −0.91 | −0.00 | −0.01 | 1.00 |
| Language_other | −0.82 | −0.82 | 0.00 | 0.00 | 1.00 |
| language_french | −0.40 | −0.34 | −0.06 | −1.92 | 0.51 |
| language_italian | −1.17 | −1.14 | −0.03 | −0.57 | 0.86 |
| language_lux | 1.30 | 1.40 | −0.10 | −3.46 | 0.12 |
| language_not_indicated | −0.70 | −0.61 | −0.09 | −2.85 | 0.27 |
| language_not_old_enough | 0.55 | 0.64 | −0.08 | −0.00 | 1.00 |
| language_portuguese | −0.45 | −0.45 | −0.00 | −0.14 | 0.96 |
| occ__resultClerical support workers | −2.73 | −3.48 | 0.77 | 6.47 | −0.65 |

| Predictor | Synthetic | Observed | Diff | Std. Coef Diff | CI Overlap |
|---|---|---|---|---|---|
| occ__resultCraft and related trades workers | −3.44 | −4.10 | 0.66 | 5.66 | −0.44 |
| occ__resultElementary occupations | −2.17 | −2.83 | 0.66 | 5.65 | −0.44 |
| occ__resultManagers | −3.85 | −4.46 | 0.61 | 5.11 | −0.30 |
| occ__resultNot applicable | −22.60 | −23.35 | 0.74 | 0.02 | 0.99 |
| occ__resultNot stated | −10.69 | −11.41 | 0.72 | 1.57 | 0.60 |
| occ__resultPlant and machine operators and assemblers | −3.43 | −4.17 | 0.74 | 6.20 | −0.58 |
| occ__resultProfessionals | −2.26 | −2.95 | 0.69 | 5.94 | −0.52 |
| occ__resultService and sales workers | −2.32 | −2.94 | 0.62 | 5.31 | −0.35 |
| occ__resultSkilled agricultural, forestry, and fishery workers | −2.70 | −3.33 | 0.63 | 5.22 | −0.33 |
| occ__resultTechnicians and associate professionals | −1.98 | −2.71 | 0.73 | 6.24 | −0.59 |

## Acronyms and Abbreviations

| Abbreviation | Meaning |
|---|---|
| **UNECE** | United Nations Economic Commission for Europe |
| **CART** | Classification and Regression Trees |
| **NSOs** | National Statistical Offices |
| **FCS** | Fully Conditional Specification |
| **GANS** | Generative Adversarial Networks |
| **pMSE** | Propensity mean squared error |
| **S_pMSE** | Standardized Propensity Mean Squared Error |
| **df** | Degrees of freedom |